

Quiz #18

The trapezoidal rule approximates the value of a definite integral. The rule is defined as:

$$\int_{x_0}^{x_1} f(x)dx = \frac{h}{2} [f(x_0) + 2f(x_0 + h) + 2f(x_0 + 2h) + 2f(x_0 + 3h) + f(x_1)]$$

$$\text{where } h = (x_1 - x_0)/4$$

Use anonymous functions to approximate the following integral:

$$\int_1^2 x^2 dx$$

Do this two ways:

1. Hard code in the function and limits. The result you should get is:

The estimate of the integral is 2.3437500

2. Change the code so that the user inputs the upper/lower limits and the function. The function will be input as a string. You should get the same result as above.

